

Supplementary Materials: Validating Experiments in Human-Robot Interaction: A State-of-the-Art Assessment

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1 Supplementary Materials Overview

This supplementary document provides additional methodological information supporting the HRI Experiment Validation Framework and systematic assessment reported in the main manuscript. It contains four components. First, Section 2 provides additional conceptual detail on the development and seven dimensions of the framework. Second, Section 3 provides the complete codebook, including variable definitions, operational coding criteria, and decision rules used in data extraction. Third, Section 4 provides additional details on papers involving multiple study settings or robot interaction modalities, as well as cases that could not be assigned to a single category in the main manuscript.

The repo also contains the coding dataset and analysis code used to generate the reported results.

- **Coding dataset (.csv):** Contains the coded methodological indicators and descriptive variables for the 73 papers included in the 2026 analysis.
- **Analysis script (.Rmd):** Contains the R code used to process the coded dataset and generate the descriptive statistics reported in the manuscript.

2 Detailed HRI Experiment Validation Framework

2.1 Framework Scope and Development

The HRI Experiment Validation Framework was developed to provide a structured basis for evaluating methodological rigor in quantitative experimental HRI research. As described in the main manuscript, the framework builds primarily on Hoffman and Zhao's recommendations for conducting HRI experiments [14] and integrates methodological principles from the broader literature on quantitative experimental research and validation [4, 10, 18, 21].

Whereas Hoffman and Zhao [14] organize methodological recommendations according to stages of the experimental process, such as developing research questions and hypotheses, designing

the experiment, sampling, conducting statistical tests, and reporting results, our framework integrates these practices within dimensions of experimental rigor. This organization allows practices occurring at different stages of an experiment to be considered in relation to the validity concerns they are intended to address.

The resulting framework comprises seven dimensions: theoretical grounding, internal validity, instrumentation validity, manipulation validity, external validity, statistical conclusion validity, and research transparency. The dimensions are not intended to represent mutually exclusive stages of an experiment. Rather, they capture complementary aspects of methodological rigor that collectively affect the extent to which conclusions from quantitative HRI experiments can be evaluated and supported.

Following Dennis and Valacich [10], the framework adopts a positivist perspective on quantitative experimental research. Accordingly, it emphasizes explicit relationships among constructs, systematic manipulation and measurement, and empirical evidence used to evaluate proposed relationships. The framework is not intended as a universal standard for qualitative, interpretive, design-oriented, or other research traditions that pursue different forms of knowledge and may require different evaluative criteria.

2.2 Theoretical Grounding

Theoretical grounding concerns whether relationships examined in an experiment are motivated by a priori theoretical or logical reasoning. Statistical relationships alone establish neither why an experimental manipulation should affect an outcome nor the conditions under which a relationship should be expected to occur. Explicit theoretical reasoning therefore helps connect individual experimental findings to a broader cumulative understanding of human-robot interaction.

Theoretical grounding does not require every experiment to invoke an established or named "Big T" theory. Dennis and Valacich [10] distinguish such theories from more focused "little t" theoretical explanations consisting of clearly specified constructs, proposed relationships among those constructs, and logical explanations for why those relationships should occur. We therefore adopt a broad definition of theoretical grounding that recognizes either established theoretical frameworks or explicit logical arguments supporting proposed relationships.

A priori specification is also relevant because explanations developed after observing results can be difficult to distinguish from predictions made before empirical testing, a practice associated with Hypothesizing After the Results are Known (HARKing) [15]. Accordingly, the framework examines whether explicit hypotheses are stated, whether hypothesized relationships are supported by theoretical or logical reasoning, and whether hypotheses appear before the Results section.

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2.3 Internal Validity

Internal validity concerns whether observed differences in an outcome can reasonably be attributed to the experimental manipulation rather than to plausible rival explanations [5, 21]. Experimental designs strengthen internal validity by minimizing systematic differences between conditions other than the manipulated variable.

Random assignment is a primary mechanism for reducing systematic pre-existing differences between experimental groups in between-subject designs. In within-subject designs, the repeated exposure of participants to multiple experimental conditions introduces potential order and carryover effects. Randomization or counterbalancing of condition order can reduce systematic confounding associated with these effects [10, 14].

Our assessment therefore focuses on whether between-subject experiments report random assignment to experimental conditions and whether within-subject experiments report randomization or counterbalancing of condition order. Although procedural standardization is also important for internal validity, it was not coded because the level of procedural detail provided in published papers did not permit this practice to be assessed consistently.

2.4 Instrumentation Validity

Instrumentation validity concerns whether measures adequately and reliably represent the constructs they are intended to capture. This issue is particularly important in HRI because many frequently examined outcomes, including trust, anthropomorphism, perceived safety, acceptance, and related psychological constructs, are latent and cannot be directly observed.

The framework considers four features of subjective measurement. First, multi-item measures can provide more reliable representations of latent constructs than single-item measures and enable the empirical assessment of measurement reliability. Second, existing or adapted measures can draw on prior theoretical development and psychometric evidence associated with established instruments. Their use does not itself guarantee validity in a new population or context, but provides a basis on which measurement can build.

Third, reliability evidence provides information regarding the consistency of measurement. Depending on the measurement design, this may include internal consistency [7], test–retest reliability [11], inter-rater reliability, or other appropriate evidence. Fourth, construct validity concerns whether an instrument captures the theoretical construct it is intended to represent. Evidence may include convergent validity, discriminant validity, factorial validity, or other analyses explicitly used to establish construct validity [6, 20].

Because these criteria apply primarily to subjective constructs represented through measurement instruments, the instrumentation-validity indicators were evaluated among papers containing at least one subjective dependent variable.

2.5 Manipulation Validity

Manipulation validity concerns whether an experimental manipulation produces the intended distinction between experimental conditions. Evidence regarding the manipulation can help establish that the independent variable was instantiated as intended before differences in downstream outcomes are interpreted [4, 20].

The appropriate form of evidence depends on the nature of the manipulation. For objectively defined manipulations, a factual check may establish whether participants noticed or correctly identified the manipulated distinction. For manipulations intended to alter a subjective perception or psychological state, a perceptual measure of the manipulated construct may provide relevant evidence [12].

At the same time, the methodological role of manipulation checks depends on the experimental design and causal claim. Manipulation checks can differ in timing, reactivity, and theoretical function, and a measure of a proposed mechanism is not necessarily equivalent to a simple treatment-fidelity check. We therefore adopted a deliberately broad and descriptive coding criterion rather than judging whether a particular form of manipulation validation was necessary for each experiment. A paper was coded as reporting a manipulation check when it explicitly reported either a factual or perceptual check of an experimental manipulation. Attention checks alone were not treated as manipulation checks.

2.6 External Validity

External validity concerns the extent to which experimental findings may generalize beyond the particular participants and contexts examined in a study. In HRI, potential boundaries on generalizability may arise from both the participant population and characteristics of the experimental context [14].

Rather than assigning each study a global judgment of external validity, we characterize observable features relevant to understanding the scope of the evidence. These include study setting, the modality through which participants encountered the robot, and the reporting of participant demographic and geographic information. We also examine whether authors explicitly discuss sample-related limitations to the generalizability of their findings.

This approach recognizes that no sample or experimental setting is inherently generalizable. The relevance of a particular population or context depends on the population and settings to which researchers intend their conclusions to apply. The coded variables therefore characterize information that allows readers to understand the empirical context and evaluate potential boundaries of generalization rather than assigning studies a binary judgment of external validity.

2.7 Statistical Conclusion Validity

Statistical conclusion validity concerns whether the quantitative evidence provides an adequate basis for drawing conclusions about relationships among the variables examined. Statistical conclusions depend not only on whether an analysis produces a statistically significant result, but also on whether the sample and analytical procedures are appropriate for the intended inference [10, 18].

We focus on three methodological practices. First, sample-size justification may include an a priori power analysis, a sensitivity analysis, or another explicit rationale for selecting the sample size, such as evidence from a pilot study or reference to prior similar studies. These forms of justification provide different information about sample-size decisions, and the coding records their presence without treating any single form as proof of analytical adequacy. Second, statistical procedures rely on assumptions or model-quality criteria that differ across analytical approaches. Reporting relevant

diagnostic procedures provides evidence that the analytical method was evaluated for appropriateness to the data and model. Third, effect sizes provide information about the magnitude of observed relationships or differences beyond statistical significance alone.

The framework does not prescribe a single statistical procedure or universal set of assumptions. Instead, coding focuses on whether papers report evidence appropriate to the analyses they employ.

2.8 Research Transparency

Research transparency concerns whether key methodological decisions and procedures are documented in ways that allow readers to understand and evaluate how an experiment was conducted. Transparent documentation also facilitates replication and cumulative scientific evaluation.

We examine two indicators in this dimension. Preregistration provides prospective documentation of research plans and analytical decisions and can help distinguish decisions specified before data analysis from those made subsequently. Institutional ethics or IRB approval provides documented evidence of institutional oversight of research involving human participants. Our coding concerns whether these practices were reported in the published paper or associated materials rather than making claims about whether an unreported practice was performed.

3 Codebook and Operational Criteria

3.1 General Coding Rules

Data extraction was conducted at the paper level. A methodological indicator was coded as present if the corresponding practice was explicitly reported for at least one applicable instance within a paper. For example, a paper was coded as reporting reliability if reliability evidence was provided for at least one applicable subjective measure. Similarly, a paper containing multiple manipulations was coded as reporting a manipulation check if at least one manipulation was explicitly checked.

This coding strategy was selected to characterize how widely methodological practices are represented across published HRI papers while giving each publication equal weight. It should therefore be interpreted as a paper-level measure of reported methodological practice rather than as an estimate of the proportion of all individual measures, manipulations, analyses, or experiments for which a practice was performed.

Coding was based on explicit evidence available in the published paper and associated supplementary materials when available. Coders did not infer that a practice had been conducted when it was not reported. Conversely, absence of reported evidence was coded as absence of reporting and should not be interpreted as definitive evidence that the practice was not performed.

The coding criteria were designed wherever possible to capture observable features of published reports rather than subjective judgments of methodological quality. For binary variables, papers were coded as *Yes* when the specified evidence was present and *No* when it was absent. Variables with additional applicability conditions or categorical response options are described below.

3.2 General Paper Characteristics

3.2.1 Publication Venue.

Definition. The publication venue of the included paper.

Coding values.

- HRI Conference
- THRI

3.2.2 Technical vs. Non-Technical Paper.

Definition. Whether the paper proposed a new robotic system, design, algorithm, or computational model and the subsequent human-subject experiment primarily evaluated that contribution.

Code Technical if. A new system, design, algorithm, or computational model was proposed and the subsequent experiment primarily evaluated that technical contribution. Thus, design-, technical-, or systems-oriented papers meeting this criterion were coded Technical (Technical_System_Paper = Yes).

Code Non-technical if. The paper did not introduce such a technical contribution, or its primary experimental contribution concerned human responses or relationships among HRI constructs. Merely using a robot or existing system in an experiment did not qualify a paper as technical.

Coding values.

- Technical
- Non-technical

3.2.3 Sample Size.

Definition. The analyzed sample size reported for each eligible experiment.

The final sample after exclusions was recorded when specified rather than the initial recruitment count. If only one sample size was reported, that value was recorded. When a paper contained multiple eligible experiments with distinct participant samples, participant counts were summed to obtain the paper-level total.

3.3 Theoretical Grounding

3.3.1 Explicit Hypothesis.

Definition. Whether the paper explicitly states at least one hypothesis regarding an expected relationship or difference.

Code Yes if.

- The paper explicitly labels a prediction as a hypothesis (e.g., H1, H2).
- The paper otherwise explicitly identifies a statement as a hypothesis even if hypotheses are not numbered.

Code No if.

- The paper reports only research questions.
- The paper discusses possible relationships or expectations without explicitly formulating them as hypotheses.
- No hypotheses are stated.

3.3.2 Theoretical or Logical Rationale for Hypotheses.

Applicability. This indicator was coded only for papers containing at least one explicit hypothesis.

Definition. Whether the paper provides an explicit theoretical or logical explanation for why at least one hypothesized relationship is expected to occur.

Code Yes if.

- At least one hypothesis is supported by an established theoretical framework.
- At least one hypothesis is supported by an explicit logical argument explaining why the proposed relationship should occur.

Code No if.

- Hypotheses are stated without an explanation for why the proposed relationship is expected.
- The paper provides only general background or cites prior findings without articulating theoretical or logical reasoning connecting the constructs.

The coding criterion concerns the presence of theoretical or logical reasoning rather than the perceived strength, novelty, or quality of that reasoning.

Coders examined both text accompanying the hypotheses and relevant earlier sections, including Related Work, Background, and Literature Review. A rationale did not have to appear immediately next to a hypothesis, but its connection to the predicted relationship had to be explicit.

The original categorical field, *Hypothesis_Type*, distinguished *Theory-driven hypotheses*, *Hypotheses without theory*, and *No hypotheses*. Under the paper-level presence rule used here, these correspond respectively to: (1) at least one explicit hypothesis with a theoretical or explanatory logical rationale; (2) explicit hypotheses but no such rationale for any hypothesis; and (3) no explicit hypotheses. A named theory was not required.

3.3.3 Hypothesis Stated Before the Results Section.

Applicability. This indicator was coded only for papers containing at least one explicit hypothesis.

Definition. Whether at least one explicit hypothesis is stated before the paper's Results section or equivalent presentation of empirical findings.

Code Yes if. The hypothesis appears before the Results section.

Code No if. The hypothesis first appears in the Results, Discussion, or another section following presentation of the empirical findings.

This indicator concerns the location of the hypothesis in the published report and does not establish when the hypothesis was originally formulated by the researchers.

3.4 Internal Validity

3.4.1 Random Assignment or Order Control.

Definition. Whether the paper reports an appropriate randomization or order-control procedure for its experimental design.

Between-subject designs. Code Yes if participants are explicitly reported as being randomly assigned or randomly allocated to experimental conditions.

Within-subject designs. Code Yes if the order of experimental conditions is explicitly reported as randomized, counterbalanced, or otherwise systematically varied to control order effects.

Mixed designs. Code Yes if the relevant between-subject assignment and/or within-subject order-control procedure is explicitly reported.

Code No if.

- Assignment or condition order is not described as randomized or controlled.
- The paper merely reports that participants were "assigned" to conditions without indicating random assignment.
- A within-subject study presents conditions in a fixed order without reported counterbalancing or randomization.

3.5 Instrumentation Validity

3.5.1 Presence of a Subjective Dependent Variable.

Definition. Whether the paper includes at least one dependent variable based on participants' subjective self-report, perception, attitude, judgment, or evaluation.

Code Yes if. At least one outcome is measured using a subjective participant response, such as a questionnaire, rating scale, or self-reported evaluation.

Code No if. All dependent variables are objective or behavioral measures, such as task completion time, error rate, physical movement, or system performance, without a subjective dependent measure.

The remaining instrumentation-validity indicators were evaluated only among papers coded Yes for this variable.

3.5.2 Multi-Item Measure.

Applicability. Papers containing at least one subjective dependent variable.

Definition. Whether at least one subjective construct is assessed using two or more items intended to represent the same construct.

Code Yes if. At least one subjective dependent construct is measured using multiple questionnaire or scale items.

Code No if. All subjective dependent constructs are assessed using single-item measures.

For example, one question answered on a seven-point Likert scale is a single-item measure: response options are not separate items. Several questions measuring different constructs do not constitute a multi-item measure of one construct.

3.5.3 Existing or Adapted Measure.

Applicability. Papers containing at least one subjective dependent variable.

Definition. Whether at least one subjective dependent measure is explicitly described as existing, previously established, validated, or adapted from prior research.

Code Yes if.

- The paper uses a previously published scale or instrument.
- The paper explicitly states that a measure was adapted or modified from a previously published instrument.

Code No if.

- All subjective measures appear to be developed specifically for the study without reference to an existing instrument.
- No information is provided regarding the origin of the measures.

Use of an existing or adapted measure was coded based on the authors' description and was not interpreted as evidence that the measure was necessarily valid in the present study.

3.5.4 Reliability Evidence.

Applicability. Papers containing at least one subjective dependent variable.

Definition. Whether the paper reports quantitative evidence explicitly used to establish the reliability or consistency of at least one subjective measure.

Code Yes if. The paper reports at least one recognized form of reliability evidence, including, for example:

- internal consistency, such as Cronbach's α ;
- composite reliability;
- test-retest reliability;
- split-half reliability;
- inter-rater reliability when applicable; or
- another quantitative statistic explicitly used as evidence of measurement reliability.

Code No if.

- No reliability evidence is reported for any subjective dependent measure.
- The paper merely cites prior reliability evidence for an instrument without reporting reliability evidence for its use in the present study.
- Correlations among substantive study variables are reported but are not used as reliability evidence.

3.5.5 Construct Validity Evidence.

Applicability. Papers containing at least one subjective dependent variable.

Definition. Whether the paper reports empirical evidence explicitly used to establish the construct validity of at least one subjective measure.

Code Yes if. The paper reports evidence such as:

- convergent validity;
- discriminant validity;
- factorial or structural validity;
- nomological validity; or
- another empirical analysis explicitly presented as evidence that the measure captures the intended construct.

Code No if.

- No construct validity evidence is reported.
- Reliability evidence is reported without construct validity evidence.
- The paper uses an established instrument but provides no construct validity evidence for the measure in the present study.

Reliability alone was not treated as evidence of construct validity.

Examples include exploratory or confirmatory factor analysis explicitly used to evaluate the measurement structure, average variance extracted (AVE) used to assess convergent validity, and multitrait-multimethod (MTMM) analysis. For instance, a paper reporting factor loadings to assess the intended structure of a subjective scale and AVE-based analyses of convergent or discriminant validity would be coded Yes. Reporting substantive correlations without an explicit construct-validation purpose was insufficient.

The measurement indicators above use the same paper-level presence rule: at least one applicable measure with the specified evidence qualifies as Yes. They do not classify papers as High, Moderate, or Low. For Reliability Evidence, reporting a reliability estimate qualifies regardless of whether it exceeds a conventional adequacy threshold such as Cronbach's $\alpha \geq .70$. Test-retest reliability refers to consistency across repeated measurements under suitably stable conditions; a pretest-posttest comparison alone does not establish it. Inter-rater or intercoder reliability counts only when it pertains to an applicable subjective dependent measure.

3.6 Manipulation Validity**3.6.1 Manipulation Check.**

Definition. Whether the paper reports at least one explicit check of an experimental manipulation.

Code Yes if. At least one manipulation is accompanied by:

- a factual check assessing whether participants recognized or identified the manipulated distinction; or
- a perceptual check assessing whether experimental conditions differed on the intended perceived construct.

Code No if.

- No manipulation check is reported.
- The paper only describes how the manipulation was implemented.
- The paper reports an attention check unrelated to whether the intended experimental manipulation was instantiated.

The coding did not evaluate whether the reported manipulation check was necessary, optimally designed, or statistically successful. It records only whether an explicit manipulation check was reported.

3.7 External Validity

External validity was characterized using both categorical descriptors of the experimental context and indicators describing the reporting of participant characteristics and generalizability limitations.

3.7.1 Study Setting.

Definition. The primary environment in which participants completed the experiment.

Coding categories.

- **Laboratory:** The experiment was conducted in a controlled laboratory or research facility.
- **Field:** The experiment was conducted in a naturally occurring or applied environment outside a conventional research laboratory.
- **Online:** Participants completed the experiment remotely through an online platform or web-based study. Examples include studies completed online through Prolific, CloudResearch, or Amazon Mechanical Turk.
- **Multiple:** The paper contained eligible experiments conducted in more than one of the above settings.
- **Not reported:** The study setting could not be determined from the published report.

Study setting was determined by where participants actually completed the experiment, rather than by the recruitment channel alone. Recruiting through an online platform did not make an experiment online if participants subsequently attended a laboratory. Field settings included real-world locations in which the studied activity took place. The original notes also allowed an *Other* category for a reported setting outside the listed types; this is distinct from *Multiple* and *Not reported*.

3.7.2 Robot Interaction Modality.

Definition. The primary modality through which participants encountered or interacted with the robot during the experiment.

Coding categories.

- **Physical robot:** Participants directly encountered or interacted with a physically embodied robot.
- **Video/image/text-based:** Participants evaluated or interacted with a representation of a robot presented through video, images, written descriptions, or related non-immersive media rather than direct physical interaction.
- **AR/VR:** Participants encountered a robot through augmented-reality or virtual-reality technology.
- **Multiple:** The paper contained eligible experiments or study components involving more than one interaction modality and could not be represented by a single category.

3.7.3 Gender Composition Reported.

Definition. Whether the paper reports the gender composition of the participant sample.

Code Yes if. The paper provides counts, percentages, or other sufficiently specific information describing participant gender.

Code No if. No participant gender information is reported.

When reported, the numbers or percentages of female, male, and other gender categories were extracted as descriptive information.

3.7.4 Mean Age Reported.

Definition. Whether the paper reports the mean age of the participant sample.

Code Yes if. A numerical mean participant age is explicitly reported.

Code No if.

- No age information is reported.
- Only an age range is reported.
- Only a median or categorical age distribution is reported without a mean.

When available, the reported mean age was extracted for descriptive analysis.

3.7.5 Participant Country Reported.

Definition. Whether the paper explicitly identifies the country or countries from which participants were recruited or in which the participant sample was located.

Code Yes if. At least one specific participant country is explicitly stated.

Code No if.

- No participant country is explicitly reported.
- Country is inferred only from author affiliations, language, recruitment platform, institutional location, or study context.

For multinational samples, all explicitly reported participant countries were recorded when available.

3.7.6 Sample-Related Generalizability Limitation.

Definition. Whether the paper explicitly discusses a limitation concerning the generalizability of its findings arising from characteristics of the participant sample.

Code Yes if. The authors explicitly identify a characteristic of their participant sample as limiting or potentially limiting the generalizability or external validity of the findings.

Code No if.

- No sample-related generalizability limitation is discussed.
- The paper discusses general limitations unrelated to the participant sample.

The coding concerns whether such a limitation was explicitly discussed rather than whether the coder judged the sample to be representative.

3.8 Statistical Conclusion Validity

3.8.1 Sample-Size Justification.

Definition. Whether the paper reports at least one explicit justification for the selected sample size.

Code Yes if. At least one eligible experiment is accompanied by:

- an a priori power analysis or sample-size calculation;
- a sensitivity analysis evaluating the effect sizes detectable with the achieved sample; or
- another explicit rationale for the selected sample size, such as evidence from a pilot study or reference to prior similar studies.

Code No if. The paper reports only the achieved sample size without a power analysis, sensitivity analysis, or another explicit rationale for selecting that sample size.

Following the paper-level presence rule, one qualifying justification is sufficient for Yes; justification of every experiment in a paper is not required. This indicator records whether a sample-size justification is reported, not whether the sample was demonstrably adequately powered.

3.8.2 Primary Statistical Analysis Approach.

Definition. The primary statistical approach used in the paper.

Coding categories.

- **Classical parametric tests:** For example, t -tests, ANOVA, ANCOVA, linear regression, and linear mixed models.
- **Non-parametric tests:** For example, Mann–Whitney U , Wilcoxon signed-rank, Kruskal–Wallis, Friedman, and Spearman rank correlation. The original scheme also grouped chi-square tests in this category.
- **SEM:** Structural equation modeling.
- **Other:** A primary approach not captured by these categories.

This field describes the analysis approach rather than its methodological quality.

3.8.3 Statistical Assumption or Model-Quality Check.

Definition. Whether the paper reports checking at least one statistical assumption or model-quality criterion relevant to an analysis used in the study.

Code Yes if. The paper explicitly reports at least one applicable diagnostic or criterion, such as:

- normality;
- homogeneity of variance;
- sphericity;
- multicollinearity;
- residual diagnostics;
- model fit or model-quality criteria; or
- another assumption or diagnostic explicitly relevant to the statistical model employed.

Code No if. No statistical assumption check or model-quality criterion is reported.

This indicator records whether at least one relevant check was reported and does not imply that every assumption associated with every analysis in the paper was examined.

Analysis-specific examples. The original notes provided the following examples to help identify relevant checks. These are illustrative, not a requirement that every listed check be reported to receive Yes:

- **Independent-samples t -test or ANOVA:** Independence, normality of model errors, and homogeneity of variance where required by the chosen procedure.
- **Repeated-measures ANOVA:** Relevant normality assumptions and sphericity when applicable.
- **ANCOVA:** Independence, residual normality, homogeneity of variance, linearity of the covariate–outcome relationship,

and homogeneity of regression slopes for a common-slope model.

- **Linear or multiple regression:** Linearity, independence of errors, residual normality when relevant to inference, homoscedasticity, and absence of problematic multicollinearity where applicable.
- **SEM:** Measurement-model reliability and validity, as well as relevant model-fit or model-quality criteria.

A reported check counts even if it detects a violation; naming an analysis alone does not demonstrate that its assumptions were checked.

3.8.4 Effect-Size Reporting.

Definition. Whether the paper reports at least one quantitative measure of effect magnitude.

Code Yes if. The paper reports an effect-size statistic, such as Cohen's d , η^2 , partial η^2 , R^2 , a standardized regression coefficient (β), Cramér's V , an odds ratio, or a correlation coefficient (r). Following the original coding notes, an explicitly reported standardized coefficient qualifies without requiring the authors to label it an effect size.

Code No if. The paper reports only p -values, test statistics (e.g., t , F , or χ^2), or raw means and standard deviations, without a separate effect-size statistic.

For example, $t(58) = 3.42$, $p = .001$, $d = 0.89$ qualifies as Yes because Cohen's d is reported. In contrast, $F(1, 108) = 3.93$, $p < .05$ without an accompanying effect size does not qualify.

3.9 Research Transparency

3.9.1 Preregistration.

Definition. Whether the paper reports that the study, hypotheses, design, or analysis plan was preregistered before the relevant empirical analysis.

Code Yes if. The paper explicitly states that the study or analysis was preregistered or provides a link or reference to a preregistration.

Code No if. No preregistration is reported.

Registration of materials, data, or a repository without prospective registration of the study or analysis plan was not coded as preregistration.

3.9.2 IRB or Ethics Approval.

Definition. Whether the paper explicitly reports institutional ethics review or approval for the human-subject research.

Code Yes if. The paper states that the research was reviewed or approved by an Institutional Review Board, ethics committee, research ethics board, or equivalent institutional body.

Code No if. No institutional ethics review or approval is reported. Statements concerning informed consent alone were not treated as evidence of institutional ethics or IRB approval unless institutional review was also explicitly reported.

4 Additional Details for Table Categories

For study setting, five papers involved multiple settings and one did not report the study setting. Specifically, [2] included both online and laboratory experiments; [19] included an online experiment and another experiment for which the setting was not reported; [16] included both online and laboratory experiments; [8] included both online and laboratory experiments; and [1] included both laboratory and field experiments. [13] did not report the study setting.

For robot interaction modality, four papers could not be assigned to a single category and were therefore excluded from the single-category breakdown reported in the main manuscript. [3] combined AR/VR and physical robot interaction; [17] combined video/description-based and physical robot interaction; [?] combined a video/description-based experiment with another experiment whose robot-exposure modality was unclear; and [9] combined an image-based experiment with an audio-guided physical prototyping task.

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